# CS3MI3 Assignment 2

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# Question 1

First, we will represent the general if-else rule:

$$\frac{t_0 \to t_0'}{\text{If } t_0 \text{ then } t_1 \to \text{If } t_0' \text{ then } t_1}$$

Then we will represent the rules for True:

$$\frac{}{\text{If True then } t_1 \to t_1}$$

and False:

$$\frac{}{\text{If False then } t_1 \to t_1}$$

# Question 2

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-- RTerm
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data RTerm
    = RTrue
    | RFalse
    | If RTerm RTerm
    deriving (Show, Eq)

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-- evalOne
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evalOne :: RTerm -> Maybe RTerm

-- Eval true/false does nothing
evalOne RTrue = Nothing
evalOne RFalse = Nothing

-- Eval with if true/false
evalOne (If RTrue t) = Just t
evalOne (If RFalse t) = Just RTrue

-- Eval t0 and decide what to return
evalOne (If t0 t1) =
    case evalOne t0 of
        Just t0' -> Just (If t0' t1)
        Nothing -> Nothing

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-- evalBig
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evalBig :: RTerm -> RTerm
evalBig t = maybe t evalBig (evalOne t)
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